

Muhammad Afsah Mumtaz

AI Engineer | Machine Learning Engineer | Full-Stack Developer

Dubai, United Arab Emirates | Open to Dubai / Abu Dhabi (on-site or hybrid) and remote worldwide

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Nationality: Pakistani | Visa: UAE Golden Visa (no sponsorship required) | Notice period: Immediate

PROFESSIONAL SUMMARY

AI and Software Engineering student (BSc Hons Information Technology, Middlesex University Dubai) seeking an **AI Engineer, Machine Learning, or Software Engineering internship or junior role**. Built and shipped production AI systems end to end — a QLoRA-fine-tuned 2B large language model (LLM) that scored 82% on ARC-Challenge (outperforming Llama 3.2 3B-Instruct) on a single 4 GB GPU, and a live multi-agent LLM trading platform deployed on Vercel — plus retrieval-augmented generation (RAG) pipelines, automation, chatbots, REST APIs, and full-stack web applications in Python, PyTorch, TypeScript, and Rust. Active open-source contributor; ship fast, test thoroughly, and adapt across AI, backend, and web. Available on-site or hybrid in Dubai / Abu Dhabi and remote worldwide; UAE Golden Visa holder — no sponsorship required, immediate start.

TECHNICAL SKILLS

Languages: Python, Rust, TypeScript, JavaScript, C++, SQL, HTML, CSS
AI / Machine Learning: Machine Learning, Deep Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Fine-Tuning (LoRA / QLoRA), Model Evaluation, Model Serving, Inference Optimization, Prompt Engineering, PyTorch, Hugging Face Transformers, Ollama, GGUF
Backend & APIs: Node.js, Express, REST API design, API integration, SQLite (FTS5), MongoDB, Vector Search (HNSW), Daemons, Linux, WSL2
Frontend & Web: Next.js, React, Tailwind CSS, Three.js, React Three Fiber, WebGL, GLSL, GSAP
DevOps & Tooling: Docker, Git, GitHub Actions, CI/CD, PowerShell, CMake, Release Packaging, Cross-Platform Scripting
Practices: Agile, Version Control, Automation, Chatbot Development, Rapid Prototyping, Technical Documentation

PROJECTS

AVA — Local-First LLM Fine-Tuning & Inference Lab

Tech: Python, PyTorch, Hugging Face Transformers, PEFT / QLoRA, RAG, Ollama, GGUF

- Engineered a local-first machine learning stack for training, evaluation, distillation, and serving of large language models on a single 4 GB VRAM laptop GPU.
- Fine-tuned Qwen3.5-2B with a 42 MB QLoRA adapter (0.58% trainable parameters) to 82.0% on ARC-Challenge, 92.0% on ARC-Easy, and 59.2% on MMLU, outperforming Llama 3.2 3B-Instruct (78.6%) on ARC-Challenge.
- Built a reproducible evaluation harness spanning 17 benchmarks (16,872 tasks); training peaked at 1.81 GB VRAM in 100 minutes, with open-sourced adapters and artifacts.

Pantheon-Trades — Multi-Agent AI Trading Platform (Live: [pantheon-trades.vercel.app](#))

Tech: Next.js 14, FastAPI (Python), PostgreSQL, Redis, Solidity, Multi-LLM (Claude, Gemini, OpenAI, +8)

- Built and shipped a live full-stack platform where an 11-agent large language model council debates every prediction-market trade before execution on Polymarket — deployed on Vercel with an on-chain audit trail.
- Engineered a pluggable multi-LLM backend across 11 providers (Anthropic Claude, Google Gemini, OpenAI, Groq, DeepSeek, and more) with a four-round deliberation and blind-voting protocol, Half-Kelly position sizing, and self-calibrating agent weights.
- Validated with 714 Python and 65 Solidity tests plus a 200-market backtest, demonstrating production-grade testing discipline.

pane — Windows-Native Linux Environment Platform

Tech: Rust, WSL2, XRDP, PowerShell, Windows packaging automation

- Developed a Windows-native control surface to create, launch, repair, and manage Linux desktop environments — shipped an Arch-first MVP packaged as pane.exe with installer shortcuts and automated release packaging.
- Designed six end-to-end lifecycle flows — onboarding, reconnect, repair, reset, diagnostics, and support bundles.

MALD — Markdown Archive & Localized Daemon

Tech: Rust, SQLite FTS5, HNSW vector indexing, RAG, Ollama, Terminal UI

- Built a privacy-first, terminal-first knowledge system as a single Rust binary combining Markdown notes, local AI search, and cited RAG chat with file- and line-level references.

- Implemented hybrid full-text and semantic search with HNSW indexing, backlinks, graph views, tasks, and plugin support.

SMNTC — Semantic Motion Engine

Tech: TypeScript, Three.js, GLSL, React, Web Components, CLI tooling

- Built a deterministic visual compiler that translates semantic intent into optimized motion and shader behavior.
- Abstracted WebGL and GLSL complexity into a tokenized intent system with React bindings, Web Components, and CLI export targets.

Entangled — Full-Stack Social Networking Web Application

Tech: JavaScript, Node.js, Express, MongoDB, REST APIs, Three.js

- Built a full-stack single-page social app with 10+ integrated features — authentication, feeds, following, messaging, notifications, profiles, search, uploads, and bookmarks — over a REST API backend.
- Integrated AI features including chatbot conversation, summarization, and code explanation.

EXPERIENCE

Open-Source Contributor, HAGI (Hypercomplex AGI Research)

2026 — Present

github.com/ShmidtS/HAGI | Remote

- Contribute on the experimental branch of an open research architecture exploring Clifford-algebra reasoning in small language models, working across PyTorch, Rust, and Lean4 in a collaborative codebase.

Sep 2025 — Present

Chairman, MDX Computing Society

Middlesex University Dubai | Dubai, UAE

- Lead the university computing society across technical initiatives, events, and community direction.

Committee Chair, BCS Student Chapter

Sep 2025 — Present

BCS, The Chartered Institute for IT | Dubai, UAE

- Chair the BCS student chapter committee, planning and executing student-led technical and professional initiatives.

Student Voice Leader

Oct 2024 — Present

Middlesex University Dubai | Dubai, UAE

- Liaise between students and the programme team on teaching quality, resources, assessments, and student experience, representing consolidated cohort feedback in academic forums.

Customer Service Representative

Oct 2021 — Jan 2022

transcosmos inc. | Debrecen, Hungary

- Resolved customer issues in a structured service environment requiring communication discipline and resilience under pressure.

EDUCATION

BSc (Hons) Information Technology

Sep 2024 — Present

Middlesex University Dubai | Dubai, UAE

A Levels

Completed

Cambridge International School, Dubai | Dubai, UAE

CERTIFICATIONS

- IBM AI Engineering Professional Certificate — Coursera
- Google IT Automation with Python Professional Certificate — Coursera
- Foundations of Project Management — Google / Coursera
- Data Visualization and Dashboards with Excel and Cognos — IBM / Coursera

ADDITIONAL INFORMATION

Languages: English (Native), Urdu (Native), Arabic (Professional Working), Hungarian (Limited Working)

Work Authorization: UAE Golden Visa holder — no sponsorship required

Availability: Internship, junior, contract, freelance, and full-time roles. On-site or hybrid in Dubai / Abu Dhabi, UAE, and remote worldwide. Immediate start.